

If you ask the kernel to profile your boot it will determine what files are accessed and then sort them according to how they are stored on your hard disk. The boot time during profiling will be long but your subsequent boots should be considerably faster. Follow these steps to profile your boot:

1. At your boot screen when grub loads press [E] (edit).
2. Navigate to the entry beginning with "kernel" and press [E] again.
3. Add **profile** without quotes at the end of this line and hit [Enter].
4. Press [B] for booting.

If you want to eliminate rebooting after updating the kernel on your server you can install kexec for warm reboots.

If you do a warm reboot you can avoid waiting while your computer re-initialises its hardware and you can skip going through the bios and bootloader. Your system can simply go to a minimal runlevel and load the new kernel image into memory and come back up. Make sure not to let kexec set itself up as the default restart handler if your system is a dual boot system and you want to reboot into a different O.S often. Follow these steps to configure kexec:

First install kexec

```
$ sudo apt-get install kexec-tools
```

Then make a backup of the reboot script and edit it

```
$ sudo cp /etc/init.d/reboot{,.bkp}; sudo gedit /etc/init.d/reboot
```

Find the do_stop function which looks like this:

```
do_stop () {
...
log_action_msg "Will now restart"
reboot -d -f -i
}
```

And replace it with this:

```
do_stop () {
...
log_action_msg "Will now restart"
if [ -x /sbin/kexec ]; then
    kexec -l --append="`cat /proc/cmdline`"
--initrd=/boot/initrd.img-`ls /lib/modules | sort -nr |
head -n 1` /boot/vmlinuz-`ls /lib/modules | sort -nr |
head -n 1`

    sync
```